

Primary VPT Chain: ProducerUA -> ProZorro -> Retail

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely I(1)-like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

standardized_type=cream

retailer=silpo

promo_variant=promo_controlled

frequency=daily

Primary chain is strictly ProducerUA -> ProZorro -> Retail. Retail is estimated separately for silpo, novus, and silpo_novus. Combined rule: daily median of available silpo and novus standardized_type prices.

Table: PreTests

series	adf_level_p	kpss_level_p	adf_diff1_p	kpss_diff1_p	adf_diff2_p	kpss_diff2_p	integration_class	stability_flag	standardized_type	retailer	promo_variant	cointegration_p	pair_n_obs
producer	3.1663521310609106e-05	0.01	1.807358615624225e-13	0.1	2.250471045105603e-20	0.1	I(1)	1.0	cream	silpo	promo_controlled	nan	nan
prozorro	5.15934840442518e-22	0.1	3.6046356335359884e-13	0.1	1.1693001750830719e-11	0.09309066682789993	I(0)	0.0	cream	silpo	promo_controlled	nan	nan
retail	0.0801729662500495	0.1	0.0004821332512077595	0.1	4.135571752859827e-05	0.1	I(1)	0.0	cream	silpo	promo_controlled	nan	nan
producer_to_prozorro	nan	nan	nan	nan	nan	nan	pair	nan	cream	silpo	promo_controlled	0.20940610652878705	199.0
prozorro_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	cream	silpo	promo_controlled	0.0010433916258889959	42.0
producer_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	cream	silpo	promo_controlled	0.9728134322997266	48.0

Table: ModelCoefficients

standardized_type	retailer	promo_variant	frequency	link	model_family	y_series	x_series	n_obs	sr_coef	lr_coef	ect_coef	ect_pvalue	asymmetry_short_p	asymmetry_long_p	vecm_rank	model_status	unreliable_flag	notes	primary_chain_doc
cream	silpo	promo_controlled	daily	producer_to_prozorro	NARDL	prozorro	producer	197	3.7936802523963236	0.33453408711897387	-1.002572700947699	2.7759014458990185e-16	0.07891071636138866	0.31445927128489587	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...	
cream	silpo	promo_controlled	daily	prozorro_to_retail	ARDL	retail	prozorro	42	-0.027190870499124387	-0.2713968465337292	nan	nan	nan	nan	nan	unreliable	1	bounds_lower_p=0.09399; b... Primary chain is strictly...	
cream	silpo	promo_controlled	daily	prozorro_to_retail	NARDL	retail	prozorro	40	-0.070469572016711	-0.23542918833682522	-0.539409399232454	0.0003540689684816318	0.23639756571220905	0.05201532870115675	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...	

Table: ResidualDiagnostics

model_family	link	y_series	x_series	ljungbox_p		arch_p		jb_p	unreliable_flag
NARDL	producer_to_prozorro	prozorro	producer	0.889712370853929	0.2018078467373643	1.393192716168525e-262			0
ARDL	prozorro_to_retail	retail	prozorro	0.004919229359110526	0.5840482434798735	0.8870969508504547			1
NARDL	prozorro_to_retail	retail	prozorro	0.5530832153948084	0.6339111636110053	0.7966413904775785			0

Table: SeriesUsed

date	producer	prozorro	retail	standardized_type	retailer	promo_variant	combined_rule
2021-02-01	16.55928516245106	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-02-02	16.56413671316524	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-02-03	16.56910996285577	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-02-04	16.57420126019822	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-02-05	16.57940695386814	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-02-06	16.58472339254112	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-02-07	16.5901469248927	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-02-08	16.59567389959844	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-02-09	16.60133523107501	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-02-10	16.60715014958769	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-02-11	16.61309747758499	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-02-12	16.61915603751542	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-02-13	16.62530465182749	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-02-14	16.63152214296973	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-02-15	16.63778733339064	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-02-16	16.64495111861826	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-02-17	16.65331829474998	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-02-18	16.66204853508837	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-02-19	16.67030151293599	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-02-20	16.6772369015954	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-02-21	16.68201437436917	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-02-22	16.68379360455986	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-02-23	16.50269387396052	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-02-24	16.03564720030959	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-02-25	15.39703236082771	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-02-26	14.70122813273551	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-02-27	14.06261329325363	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-02-28	13.5955666196027	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-03-01	13.41446688900336	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-03-02	13.42360215097776	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-03-03	13.44716151080648	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-03-04	13.47937532934779	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-03-05	13.51447396745996	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-03-06	13.54668778600126	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-03-07	13.57024714582998	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-03-08	13.57938240780438	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-03-09	13.57882544089713	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-03-10	13.57724782620368	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-03-11	13.57478949276645	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...
2021-03-12	13.5715903696279	nan	nan	cream	silpo	promo_controlled	daily_median_of_available...

Table: LagProfile

	pair	lag	corr
producer_to_prozorro	1	0.06146229004493464	
producer_to_prozorro	2	0.06187172383641456	
producer_to_prozorro	3	0.06314532535711788	
producer_to_prozorro	4	0.06320312979983103	
producer_to_prozorro	5	0.06245313280573821	
producer_to_prozorro	6	0.06282849386241454	
producer_to_prozorro	7	0.06579244345091867	
producer_to_prozorro	8	0.07036064213557568	
producer_to_prozorro	9	0.0740021648388298	
producer_to_prozorro	10	0.07999426010710693	
producer_to_prozorro	11	0.08672309347422189	
producer_to_prozorro	12	0.0955548834637671	
producer_to_prozorro	13	0.10393825381312094	
producer_to_prozorro	14	0.10980154996288898	
producer_to_prozorro	15	0.11026586073085809	
producer_to_prozorro	16	0.1040630388642432	
producer_to_prozorro	17	0.09141072312581083	
producer_to_prozorro	18	0.07741161321894428	
producer_to_prozorro	19	0.06731513319302308	
producer_to_prozorro	20	0.061079129894117214	
producer_to_prozorro	21	0.05953068607471382	
producer_to_prozorro	22	0.06072146887343792	
producer_to_prozorro	23	0.05896910731704648	
producer_to_prozorro	24	0.05122603053484502	
producer_to_prozorro	25	0.040063549807138873	
producer_to_prozorro	26	0.029907058109922408	
producer_to_prozorro	27	0.021911020291762194	
producer_to_prozorro	28	0.01727374726403975	
producer_to_prozorro	29	0.015666284093778012	
producer_to_prozorro	30	0.013940599977148376	
prozorro_to_retail	1	-0.012329464894321948	
prozorro_to_retail	2	0.022282149176227548	
prozorro_to_retail	3	-0.06480592881330549	
prozorro_to_retail	4	-0.3405588807782156	
prozorro_to_retail	5	-0.11009469381741226	
prozorro_to_retail	6	-0.11991331282533414	
prozorro_to_retail	7	0.16515282202902284	
prozorro_to_retail	8	0.17560440653777376	
prozorro_to_retail	9	0.1607219357774288	
prozorro_to_retail	10	0.007638898487336116	

Table: ModelEligibility

standardized_type	retailer	promo_variant	frequency	integration_producer	integration_prozorro	integration_retail	n_obs_producer	n_obs_prozorro	n_obs_retail	n_obs_pair_prod_prozorro	n_obs_pair_prozorro_retail	any_i2	eligibility_note
cream	silpo	promo_controlled	daily	I(1)	I(0)	I(1)	1793	209	48	199	42	0	Admissible for ARDL/ECM s...

Table: NARDL_Multipliers

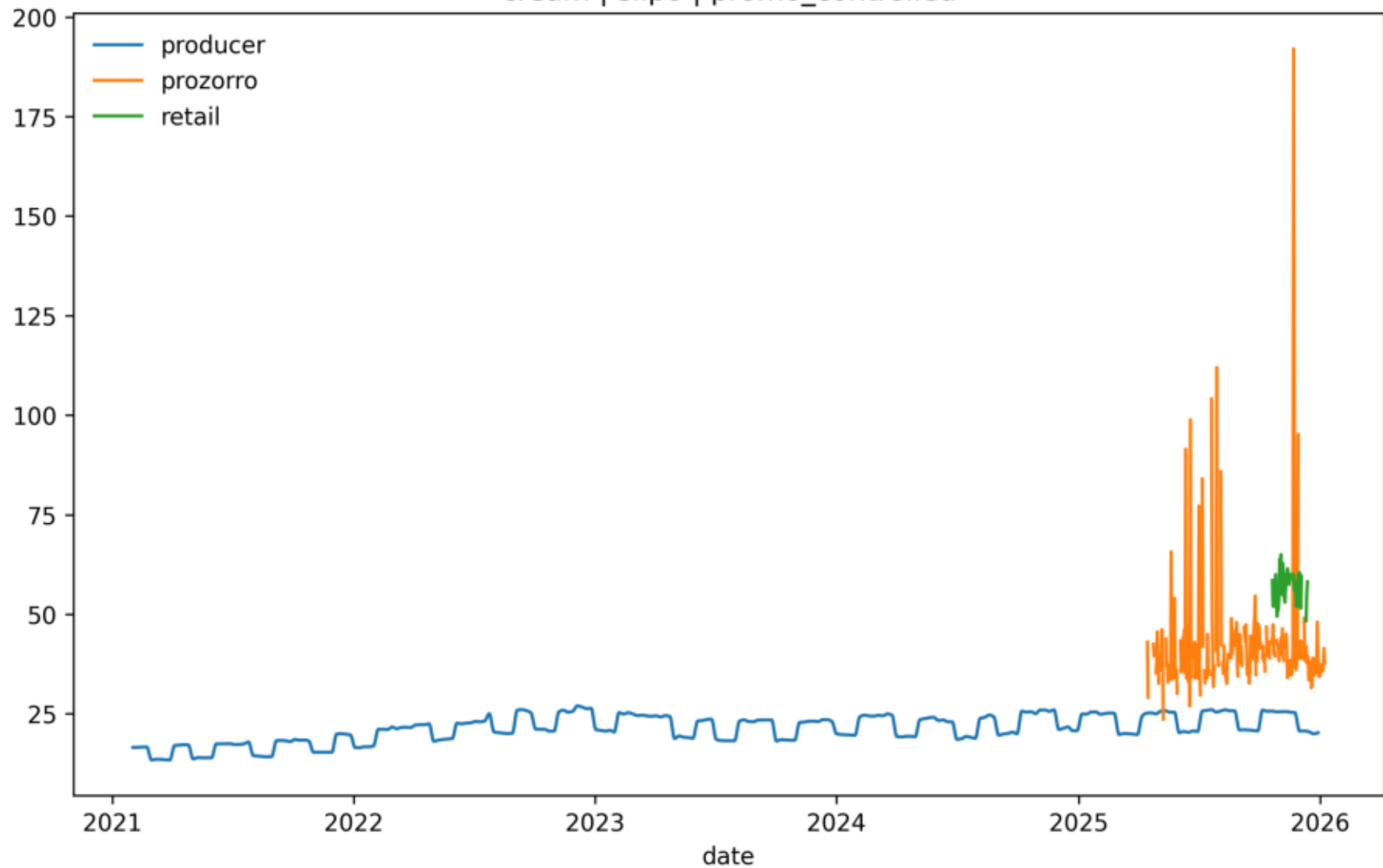
horizon	mult_pos	mult_neg	link
0	2.9056194294341315	-0.8880608229621922	producer_to_prozorro
1	3.166679443444792	-0.9678500646420563	producer_to_prozorro
2	3.1901347969659235	-0.9750188563770777	producer_to_prozorro
3	3.192242180629738	-0.9756629479094464	producer_to_prozorro
4	3.1924315218776758	-0.9757208173409214	producer_to_prozorro
5	3.1924485335442205	-0.9757260167123321	producer_to_prozorro
6	3.192450061984546	-0.975726483858153	producer_to_prozorro
7	3.192450199309706	-0.9757265258296146	producer_to_prozorro
8	3.192450211647904	-0.9757265296006076	producer_to_prozorro
9	3.1924502127564494	-0.9757265299394184	producer_to_prozorro
10	3.1924502128560484	-0.9757265299698594	producer_to_prozorro
11	3.1924502128649968	-0.9757265299725945	producer_to_prozorro
12	3.192450212865801	-0.9757265299728402	producer_to_prozorro
13	3.192450212865873	-0.9757265299728622	producer_to_prozorro
14	3.1924502128658796	-0.9757265299728642	producer_to_prozorro
15	3.19245021286588	-0.9757265299728644	producer_to_prozorro
16	3.19245021286588	-0.9757265299728645	producer_to_prozorro
17	3.19245021286588	-0.9757265299728645	producer_to_prozorro
18	3.19245021286588	-0.9757265299728645	producer_to_prozorro
19	3.19245021286588	-0.9757265299728645	producer_to_prozorro
20	3.19245021286588	-0.9757265299728645	producer_to_prozorro
0	0.0464158882449621	0.1168854602616731	prozorro_to_retail
1	0.03956155280035692	0.09962472943169626	prozorro_to_retail
2	0.040573747463943266	0.1021736591969376	prozorro_to_retail
3	0.04042427447049661	0.10179725317959507	prozorro_to_retail
4	0.040446347473421504	0.1018528378770723	prozorro_to_retail
5	0.040443087904941014	0.10184462956357229	prozorro_to_retail
6	0.04044356925252726	0.10184584170312819	prozorro_to_retail
7	0.04044349817087944	0.10184566270383644	prozorro_to_retail
8	0.04044350866766139	0.10184568913705222	prozorro_to_retail
9	0.04044350711757871	0.1018456852336016	prozorro_to_retail
10	0.0404435073464828	0.10184568581003267	prozorro_to_retail
11	0.04044350731268003	0.10184568572490985	prozorro_to_retail
12	0.04044350731767176	0.10184568573748011	prozorro_to_retail
13	0.04044350731693462	0.10184568573562383	prozorro_to_retail
14	0.04044350731704348	0.10184568573589796	prozorro_to_retail
15	0.0404435073170274	0.10184568573585748	prozorro_to_retail
16	0.040443507317029774	0.10184568573586344	prozorro_to_retail
17	0.04044350731702942	0.10184568573586257	prozorro_to_retail
18	0.040443507317029476	0.1018456857358627	prozorro_to_retail

Table: VECM_IRF

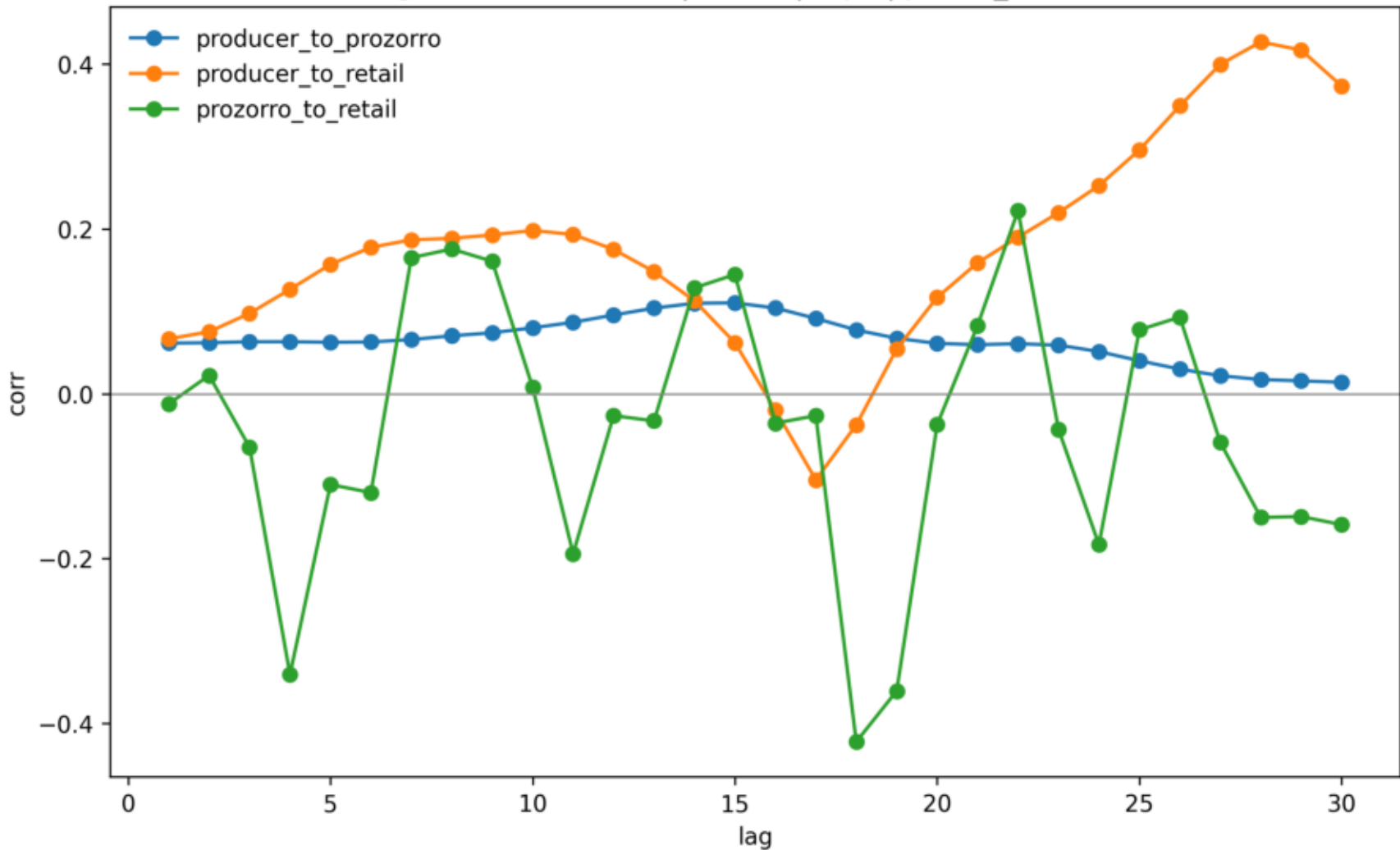
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time_series_promo_controlled.png

cream | silpo | promo_controlled



Lag-Correlation Profile | cream | silpo | promo_controlled



NARDL Dynamic Multipliers | cream | silpo | promo_controlled

